CORNERSTONE PROJECT REPORT

Tourist Destination Recommendation System Using KNN

Wayfinder: A personalised city discovery planner

Submitted for the Machine Learning Laboratory

Project	Tourist Destination Recommendation System Using KNN
Technology	Python, Pandas, scikit-learn and Streamlit
Dataset	Worldwide Travel Cities dataset - 560 city profiles across 167 countries
Algorithm	K-Nearest Neighbors with StandardScaler

Academic report prepared in the supplied Cornerstone Project Report Format.

Abstract

Wayfinder is a web-based travel recommendation system that identifies city destinations aligned with a traveller's budget, available days, region, country, and six travel-style preferences. The system uses a 560-record worldwide city dataset and converts the six user-facing preferences into nine travel-theme scores. After filtering the catalogue, StandardScaler normalises feature values and a Euclidean K-Nearest Neighbors model retrieves the five closest city profiles. The interface additionally presents a map, destination links, live exchange-rate conversion, weather information, cost-planning estimates, favourites, comparison, and export tools. This report documents the problem, data, method, implementation, diagnostic experiments, findings, limitations, and future work.

Keywords: travel recommendation, K-Nearest Neighbors, StandardScaler, Streamlit, content-based filtering.

Table of Contents

1. Problem Statement
2. Objectives
3. Introduction and Motivation
4. Literature Review / Existing Approaches
5. Proposed Methodology
6. Dataset Description
7. Data Preprocessing and Exploratory Analysis
8. Implementation Details
9. Experimental Design
10. Results and Analysis
11. Discussion
12. Conclusion and Future Scope
13. References

1. Problem Statement

Choosing a destination from a large global catalogue is difficult when travellers have several constraints at the same time. A person may have a limited budget, only a few days, a preferred region, and a strong interest in culture, nature, food, adventure, urban exploration, or wellness. Search engines return many results but do not provide a transparent similarity-based ranking across these inputs.

The project solves this discovery problem for leisure travellers. Its inputs are a total budget and currency, trip duration, region, country, and six interest ratings. Its expected output is a ranked list of up to five destination profiles, their map locations, suggested stay, cost-planning estimate, direct external links, and a measure of KNN closeness. The system supports destination exploration; it does not make bookings or claim to provide live booking prices.

2. Objectives

No.	Specific objective	Success criterion
1	Build a usable travel-planning interface.	Accept budget, days, region, country and six interests.
2	Recommend similar destinations with KNN.	Return up to five nearest valid profiles.
3	Normalise recommendation features.	Apply StandardScaler before Euclidean distance.
4	Provide useful decision support.	Show location links, map, cost estimate and suggested stay.
5	Support reusable trip planning.	Provide favourites, comparison, export and share settings.

The objectives are measurable through the implemented input controls, successful recommendation runs, the presence of scaled KNN ranking, and the rendered decision-support features.

3. Introduction and Motivation

Travel discovery is a practical recommendation problem because preferences are multidimensional. A cultural traveller and a nature traveller can have the same budget and duration but seek completely different destinations. Content-based recommendation is appropriate where a structured catalogue describes each destination and user preferences can be represented on compatible features.

The project is relevant to students and independent travellers who need an understandable starting point before consulting official tourism websites and booking providers. Wayfinder exposes the criteria used for ranking and reports the KNN distance. This transparency is useful in a classroom project because users can relate each recommendation to the preferences entered.

Potential applications include travel discovery portals, tourism information dashboards, itinerary ideation tools, and personalised content modules. A production booking product would require verified prices, availability, route-aware transport quotes, user-feedback data, and privacy controls beyond this prototype.

4. Literature Review / Existing Approaches

Recommender systems commonly use collaborative filtering, content-based filtering, or hybrid approaches. Content-based methods recommend items whose attributes resemble a user profile; this fits a destination catalogue containing explicit interest scores. Ricci, Rokach and Shapira describe tourism recommender systems as context-sensitive systems where practical travel constraints are important [1].

KNN is a simple non-parametric approach that retrieves nearby observations under a selected distance measure. Scikit-learn documents NearestNeighbors as an unsupervised neighbour search method that supports Euclidean distance [2]. Feature scaling is necessary when feature ranges differ; StandardScaler removes the mean and scales to unit variance [3].

Existing commercial travel systems often depend on interaction history, reviews, prices, and inventory. They may be more personalised at scale but are less transparent and require private platform data. This project uses published city attributes instead. Its limitations are the absence of verified live prices, route-specific costs, personal travel history, and supervised relevance labels.

Approach	Strength	Limitation for this project
Keyword/search filtering	Fast and familiar	Does not rank by a combined preference profile.
Collaborative filtering	Learns from similar users	Needs large interaction histories and raises privacy issues.
Content-based KNN	Transparent and works with catalogue data	Depends on attribute quality; may over-specialise.
Hybrid recommender	Can improve ranking quality	Needs additional behavioural and live data.

5. Proposed Methodology

The proposed solution is a content-based nearest-neighbour recommender. The traveller first supplies practical filters and six interest ratings. A weighted mapping expands those ratings into the source dataset's nine dimensions: culture, adventure, nature, beaches, nightlife, cuisine, wellness, urban, and seclusion. The system then filters eligible cities, standardises the nine dimensions, and queries a Euclidean KNN model for the five closest profiles.

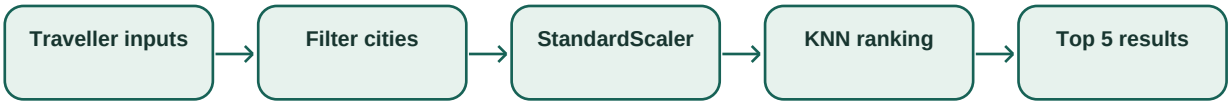

Figure 1. Overall Wayfinder recommendation workflow.

The budget input is converted to USD using a live exchange-rate service when available. The converted amount chooses the source data's Budget, Mid-range, or Luxury filter. The recommendation card displays a currency planning estimate derived from the source category and suggested duration. It is deliberately labelled as an estimate because the dataset does not contain verified booking prices.

6. Dataset Description

The main dataset is the Worldwide Travel Cities (Ratings and Climate) dataset used in the application [4]. The prepared CSV contains 560 observations, 559 distinct city names, 167 countries, seven regions, and 19 columns. Each row is one destination profile. The task is a ranking problem, so there is no target variable or class distribution.

Field group	Fields	Use in project
Identity and context	id, city, country, region, description	Display, filtering and external links.
Geographic	latitude, longitude	Interactive map and weather request.
Practical	ideal_durations, budget_level	Days and budget filtering; planning estimate.
Preference features	culture, adventure, nature, beaches, nightlife, cuisine, wellness, urban, seclusion	Standardised KNN feature matrix.
Climate metadata	avg_temp_monthly	Available source metadata; not used for KNN ranking.

Data quality checks performed by the app validate required columns and drop rows missing a KNN interest score or coordinates. The evaluated prepared data had zero missing values after this validation. City scores are publisher-provided attributes, rather than independently verified visitor ratings or live conditions.

7. Data Preprocessing and Exploratory Analysis

The application reads the CSV with Pandas, verifies the required schema, converts the nine interest columns to numeric values, removes incomplete feature rows, and casts identifiers to strings. Duration text is parsed into tags so that a traveller's selected stay can filter eligible records. No label encoding is needed because country, region, duration, and budget category are used as filters rather than KNN input features.

StandardScaler is applied after filtering and before neighbour search. This gives each interest feature zero mean and unit variance within the candidate set. It prevents a feature with a larger numeric spread from dominating Euclidean distance. The source scores share a 0-5 style scale, but scaling remains an explicit project requirement and makes the pipeline robust to variation in feature distributions.

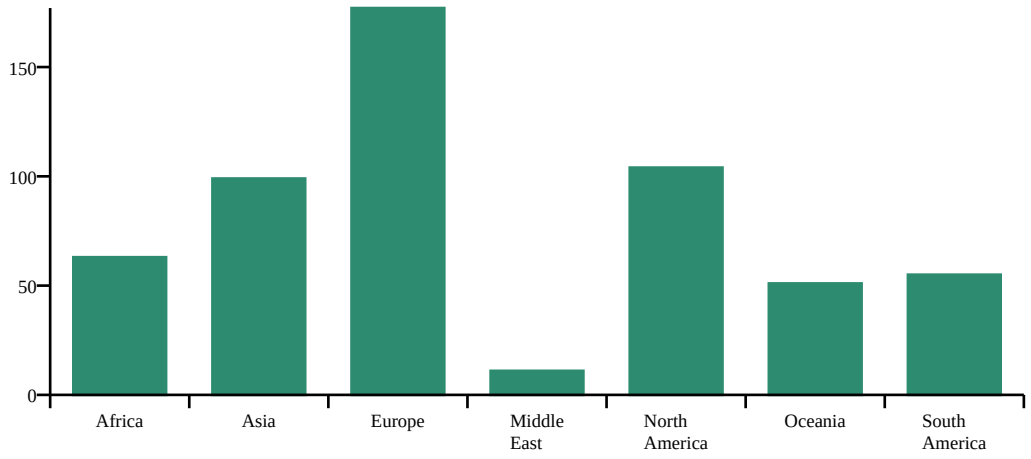

Figure 2. City profiles by source-data region (n = 560). Europe contributes the largest share; the Middle East is sparsely represented.

Budget category	Profiles	Share
Budget	145	25.9%
Mid-range	339	60.5%
Luxury	76	13.6%

Table 1. Distribution of source-data budget categories. The Mid-range category dominates the catalogue, which can affect availability after a budget filter.

8. Implementation Details

Layer	Implementation
Language	Python 3.13
Data processing	Pandas and NumPy-compatible numeric arrays
ML algorithm	scikit-learn NearestNeighbors; Euclidean metric; n_neighbors = min(5, candidate count)
Scaling	scikit-learn StandardScaler
Web interface	Streamlit
Live data	Frankfurter exchange rates (one-hour cache); Open-Meteo forecast (15-minute cache)
Exports	CSV through Streamlit; PDF through ReportLab

The KNN model is fitted after every filter selection because the candidate pool changes. This is appropriate for a small catalogue of 560 rows and keeps scaling aligned with the available candidates. The six UI sliders are mapped by weighted rules to the nine dataset dimensions; for example, Cultural Immersion contributes most strongly to culture, with smaller cuisine and urban contributions.

The cost breakdown divides a destination planning estimate into generic flight allowance, accommodation, food, and activities/local travel components. These values are a transparent heuristic for comparison only. The application also handles live-service failures by keeping recommendations available and displaying an offline exchange-rate fallback or weather availability message.

9. Experimental Design

This is an unsupervised content-based recommender, so there are no ground-truth relevance labels for train-test accuracy, precision, recall, or MSE. A conventional train-validation-test split would not measure ranking correctness without historical user ratings. Instead, the experiment uses a fixed prepared catalogue and three reproducible traveller profiles to test whether scaled KNN returns coherent nearest matches.

Setting	Value
Dataset split	No supervised split; all valid catalogue rows are available for nearest-neighbour retrieval.
Reproducibility	Fixed CSV snapshot; deterministic Euclidean nearest-neighbour search.
Candidate filters	No filter for diagnostic profiles; all 560 valid rows considered.
Model	NearestNeighbors(metric='euclidean', n_neighbors=5).
Feature preparation	StandardScaler fit on the candidate matrix before querying.
Evaluation evidence	Top-five distance, recommended locations, and qualitative alignment with the input profile.

The three profiles are: Culture seeker (high culture, cuisine, and urban preference); Nature adventurer (high nature, adventure, and seclusion preference); and Wellness traveller (high wellness, nature, and seclusion preference). The experiment records the mean top-five standardised distance and destination list. Lower distance indicates closer feature similarity, but it is not a probability or a satisfaction score.

10. Results and Analysis

All three diagnostic queries completed successfully and returned five nearest profiles. Table 2 contains results computed from the implemented scaled KNN pipeline over the 560 valid rows. The nature/adventure query obtained the lowest mean top-five distance (2.048), while the wellness query obtained the highest (4.356). This difference is evidence that the available city profiles describe some preference combinations more closely than others.

Profile	Mean top-5 distance	Nearest example	Interpretation
Culture seeker	2.299	Dhaka, Bangladesh	High culture/cuisine/urban profile returned heritage-oriented and urban cities.
Nature adventurer	2.048	Svalbard, Norway	Nature- and adventure-led cities and parks ranked closest.
Wellness traveller	4.356	Saaremaa Island, Estonia	Wellness profile had fewer close catalogue matches, indicating weaker coverage.

Table 2. KNN diagnostic results. Distances are standardised Euclidean distances; lower is a closer match.

Profile	Top five returned destinations
Culture seeker	Dhaka; Giza; Detroit; Varanasi; Phnom Penh
Nature adventurer	Svalbard; Zion National Park; Yellowstone National Park; Iquitos; Galapagos Islands
Wellness traveller	Saaremaa Island; Avarua; Nuwara Eliya; Siwa Oasis; Bora Bora

Table 3. Actual top-five destination outputs from the diagnostic profiles. The results are plausible with respect to the high-level input themes, but plausibility is not a substitute for user-study validation.

The user interface presents the ranked cards with a currency estimate, suggested days, match distance, maps link, detail link, save action, cost breakdown, map, weather panel, comparison panel, and export panel. These outputs support inspection and comparison after the KNN ranking has been produced.

11. Discussion

The implementation achieved the core objectives: it accepts the specified trip inputs, scales the preference features, applies KNN, and returns up to five matches. It also turns the ML result into a practical planner through map links, saved destinations, comparison, exports, currency conversion, and weather information.

The strongest aspect is transparency. The app makes the selected filters visible and reports a distance value rather than presenting an unexplained recommendation. The small catalogue makes response time suitable for Streamlit and permits refitting after filters change.

The primary limitation is data quality and coverage. The city dataset contains category-level budget information rather than real local prices, and the source interest scores do not represent a validated population of travellers. The planning cost is therefore a heuristic, not a quotation. Country and regional coverage are imbalanced, and the wellness test profile had the largest mean distance. Live exchange rates and weather improve utility but depend on external services. A future evaluation should collect user relevance ratings and measure ranking quality with held-out feedback.

12. Conclusion and Future Scope

Wayfinder demonstrates a complete content-based travel destination recommender using K-Nearest Neighbors. It combines practical filters with six understandable interest controls, converts them into a nine-feature representation, standardises the candidate matrix, and retrieves the five nearest city profiles. The app turns those results into a usable travel-discovery flow through currency estimates, destination links, a map, weather, comparison, favourites, exports, and shared settings.

The report's diagnostic results show that the model produces coherent destination sets for culture and nature/adventure profiles and reveals weaker catalogue coverage for the wellness profile. The project is suitable as an ML laboratory application because the method, scaling step, data limitations, and outputs are visible and reproducible.

Future work should add verified price and accommodation APIs, origin- and date-aware flight estimates, official tourism content, multilingual descriptions, destination images with reliable licenses, accessibility validation, user accounts with consent, feedback collection, hybrid recommendations using behaviour data, and offline ranking-quality studies with explicit relevance judgements.

13. References

[1] F. Ricci, L. Rokach, and B. Shapira, Recommender Systems Handbook, 3rd ed. Springer, 2022.

[2] scikit-learn developers, 'NearestNeighbors,' scikit-learn documentation. Available: https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.NearestNeighbors.html. Accessed: Oct. 14, 2026.

[3] scikit-learn developers, 'StandardScaler,' scikit-learn documentation. Available: https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html. Accessed: Oct. 14, 2026.

[4] Furkan Ima, 'Worldwide Travel Cities: Ratings and Climate,' Kaggle dataset. Available: https://www.kaggle.com/datasets/furkanima/worldwide-travel-cities-ratings-and-climate. Accessed: Oct. 14, 2026.

[5] Frankfurter, 'Foreign exchange rates API.' Available: https://frankfurter.dev/. Accessed: Oct. 14, 2026.

[6] Open-Meteo, 'Weather Forecast API.' Available: https://open-meteo.com/. Accessed: Oct. 14, 2026.

[7] Streamlit, 'Streamlit documentation.' Available: https://docs.streamlit.io/. Accessed: Oct. 14, 2026.

Appendix A - Code submission

The implementation files are app.py, worldwide_travel_cities.csv, requirements.txt, README.md, and DATA_SOURCES.md in the project repository. Long source-code listings are intentionally omitted from the main report in accordance with the supplied report guidelines.